

Circulation of railway rolling stock: a branch-and-price approach

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Abstract

In this paper, we describe a model and a branch-and-price algorithm to determine an efficient railway rolling stock circulation on a set of interacting train lines. Given the timetable and the passengers' seat demand, the model determines an allocation of rolling stock to the daily trips. In order to efficiently utilize the train units, they can be added to or removed from the trains at some stations along the lines. These changes in train compositions are subject to several constraints, mainly corresponding to the order of the train units within the trains. A solution is evaluated based on three criteria, i.e. (i) the service to the passengers, (ii) the robustness, and (iii) the cost of the circulation. The developed branch-and-price algorithm was tested on a number of real-life instances of NS Reizigers, the main Dutch operator of passenger trains, thereby outperforming the commercial solver CPLEX 8.0.

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1. Introduction

1.1. The rolling stock circulation problem

The efficient circulation of railway rolling stock is an important problem for operators of passenger trains, since the rolling stock represents a huge capital investment. The rolling stock is also an investment that cannot be changed frequently, because it usually has an economic life cycle of several decades. For these reasons, it has to be decided carefully how many carriages or train units are necessary per scheduled train in order to attain a certain quality level for the passengers.

In this paper we describe a branch-and-price approach for determining efficient circulations of a number of *train units* on a single line or on a set of interacting lines. Here, a line is defined by two endpoints between which a fixed number of trains run up and down according to the timetable. In order to obtain per trip a better match between the supply of seats and the demand for seats, the compositions of the trains can be changed at some stations by coupling or uncoupling train units. For example, train units may be uncoupled from a train after the morning rush hours, and they may be coupled again before the afternoon rush, possibly onto another train. These shunting operations, which are carried out in the short time interval between the arrival of a train at a station and its subsequent departure, are subject to several practical rules related to the feasibility of the transitions of the train compositions.

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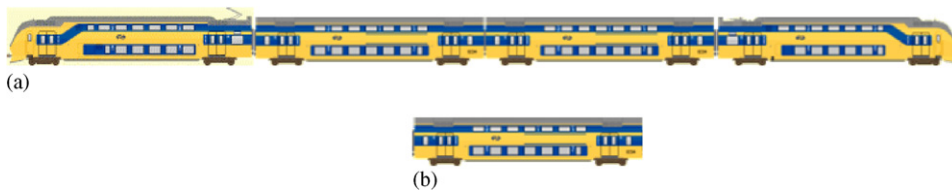


Fig. 1. (a) Double deck train unit and (b) double deck carriage.

The primary objective pursued in the planning process of a rolling stock circulation is the maximization of the service to the passengers by minimizing the expected seat shortages. Given the current timetable and the expected numbers of passengers, some seat shortages may be unavoidable, because of a limitation on the train lengths or because of a limited amount of rolling stock. Other relevant concerns are the robustness and the total cost of the rolling stock circulation.

Note that the railway system in the Netherlands is quite dense. This allows one to take into account maintenance planning and maintenance routing of train units only implicitly in the rolling stock circulations. That is, in addition to the train units that are required for operating the train services, a sufficient number of train units is reserved for maintenance. Maintenance routing is carried out in the operations by exchanging duties of individual train units [1]. Therefore, the maintenance routing problem can be ignored in the rolling stock circulation planning problem.

Although the model described in this paper is also valid for solving operational rolling stock circulation problems, the primary objective of the model is to investigate the effect of changes in some input parameters on the objectives, e.g. the number of train units allocated to a line, an additional station where units can be (un)coupled, a change in the allowable train length on some trips, etc.

Our approach is tested on several real-life instances of NS Reizigers, the main Dutch operator of passenger trains. This operator mainly uses train units to operate its train services. Train units differ from train carriages, because they can move individually in both directions without a locomotive. Fig. 1a presents a double deck train unit with four carriages and Fig. 1b presents a double deck carriage. There exists other train units of this double deck type with a different number of carriages (three carriages or six carriages), subsequently called *subtypes*. Such a train unit is indivisible and a train can be composed of several of those subtypes of train units. Different *types* of train units (e.g. double deck versus single deck), however, cannot be combined with each other in one train. It is preferable to assign only one type of train units to a line or to a set of interacting lines, since in case of disruption of the operations (e.g. a switch or electricity breakdown), the circulation will not be completely disorganized, and it will be much easier to return to the regular situation afterwards.

In this paper, we describe a *branch-and-price approach* for solving the rolling stock circulation problem. During the last decades, many hard integer programming problems have been successfully solved exactly with branch-and-price methods. The first papers on branch and price or Integer Programming column generation are [2,3] on routing problems. Savelsbergh [4] presents an efficient branch-and-price algorithm for the generalized assignment problem. For a general exposition on branch and price, we refer to Vanderbeck and Wolsey [5], Barnhart et al. [6] and Vanderbeck [7].

The rolling stock circulation problem has similarities with the multicommodity flow problem (MCFP), where the flows of different commodities must be routed through a graph from supply to demand nodes without exceeding the capacities of the arcs (see e.g. [8]). In our problem, the commodities correspond to the rolling stock subtypes, and the graph is determined by the different trips that must be served. The capacity of an arc corresponds to the maximum number of carriages that can be deployed on a trip, because of the lengths of the platforms in the stations along the line.

Column generation approaches and branch-and-price algorithms have been proposed to tackle variants of the MCFP. For example, Barnhart et al. [9] present a branch-and-price-and-cut algorithm to solve an origin–destination integer MCFP. Holmberg and Yuan [10] present a column generation approach for an MCFP with side constraints. Usually, for every commodity, paths through a network are generated, where the master problem contains arc capacity constraints. In our problem, a complicating issue is the fact that we have to take into account the *orders* of the commodities on the arcs, as will be explained later. Therefore, we generate paths corresponding to *joint flows* of several commodities that meet capacity constraints as well as certain transition constraints.

The remainder of this paper is organized as follows. Section 2 describes the rolling stock circulation problem in more detail. Section 3 presents the model and an Integer Programming formulation, that is used as a benchmark for

our branch-and-price approach. In Section 4, we present a Dantzig–Wolfe reformulation for the problem. Section 5 outlines the branch-and-price procedure. Section 6 discusses the results of our numerical experiments. Finally, we describe some conclusions and subjects for further research in Section 7.

2. Problem description

2.1. Lines

A line is defined by two endpoints between which a fixed number of trains run up and down according to the timetable. This number of trains is determined by the circulation time of the line, including the return times at the endpoints, and the line's frequency. For example, Fig. 2 presents the railway map of the Netherlands. The bold line presents the intercity 2100 line running between Amsterdam (Asd) and Vlissingen (Vs), the 2400 line between Asd and Dordrecht (Ddr), and the 2600 line between Asd and The Hague (Gvc). Roosendaal (Rsd) is a relevant underway station of the 2100 line. The lines interact with each other, because, after a 2600 train has made his journey from Asd to Gvc and back, its train number changes and it performs a roundtrip to Vs (2100) or Ddr (2400). The sequence is not arbitrary. In fact, if a 2100, 2400 or 2600 train arrives in Asd, its next journey is the first 2100, 2400 or 2600 roundtrip that leaves Asd after its arrival. The cyclic nature of the timetable results in the same sequence of roundtrips, which is presented in Table 1. The last two columns present the arrival and departure time for the trips from and to Asd. After the fourth roundtrip, a train carries out the first roundtrip again. The total time, including the waiting time, for the four roundtrips is 14 h. Therefore, at least 14 trains are required to run the timetable for the 2100, 2400 and 2600 lines. As is illustrated by Table 1, we see that after 14 h the train is back in Amsterdam and ready to carry out the first roundtrip again. Due to some deviations from the regular pattern during the morning rush hour, even 15 trains are required.

As is illustrated by the example, a train makes several consecutive trips during the day. In the dense Dutch railway timetable, the sequence of the trips is fixed and input to the model. A trip is characterized by a departure and an arrival time and by a departure and an arrival station. In both the arrival and the departure station, coupling and uncoupling can

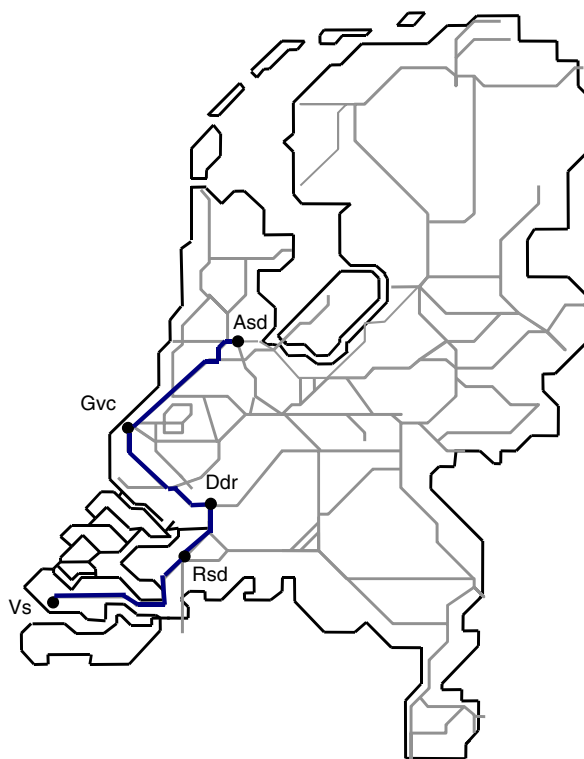


Fig. 2. Lines 2100, 2400, 2600.

Table 1
Lines 2100, 2400 and 2600

	Roundtrip	Train number	Asd–Xyz	Xyz–Asd
1	Asd–Gvc–Asd	2600	$x : 13 - (x + 1) : 03$	$(x + 1) : 29 - (x + 2) : 20$
2	Asd–Ddr–Asd	2400	$(x + 2) : 29 - (x + 3) : 47$	$(x + 4) : 13 - (x + 5) : 33$
3	Asd–Gvc–Asd	2600	$(x + 5) : 43 - (x + 6) : 33$	$(x + 6) : 59 - (x + 7) : 50$
4	Asd–Vs–Asd	2100	$(x + 7) : 59 - (x + 10) : 36$	$(x + 11) : 26 - (x + 14) : 03$

Table 2
Consecutive trips of a 2100–2400–2600 train

t_d	t_a	Nr	s_d	s_a
7:26	8:17	2122	Vs	Rsd
8:20	10:03	2122	Rsd	Asd
10:13	11:03	2641	Asd	Gvc
11:29	12:20	2630	Gvc	Asd
12:29	13:47	2449	Asd	Ddr
14:13	15:33	2444	Ddr	Asd
15:43	16:33	2663	Asd	Gvc
16:59	17:50	2652	Gvc	Asd
17:59	19:42	2171	Asd	Rsd
19:45	20:36	2171	Rsd	Vs
21:26	22:17	2178	Vs	Rsd
22:20	0:03	2178	Rsd	Asd
0:13	1:03	2697	Asd	Gvc

take place. On a trip, the train can make intermediate stops at other stations, but those are only relevant if the train composition can be changed there. In the above example, the train composition can be changed in Rsd on the trip Asd–Vs and Vs–Asd. Therefore, those trips are divided into two trips Asd–Rsd and Rsd–Vs or Vs–Rsd and Rsd–Asd. On the contrary, a stop in Rotterdam is not relevant, since a change in train composition is not allowed here. Table 2 presents the consecutive trips on a single day of one of the 15 trains. The table shows per trip the departure and arrival time, the train number, and the departure and arrival station.

Finally, for each trip, the required capacity of first and second class passengers' seats is determined as follows. Every trip is divided into the subtrips that correspond to the intermediate stops that the train makes. For example, between Asd and Gvc, the 2600 train stops at Schiphol (Shl) and Leiden (Ledn), resulting in three subtrips, i.e. Asd–Shl, Shl–Ledn, Ledn–Gvc. For these subtrips, first and second class passengers' counts of the conductors are known. Those counts are then changed into the required capacity by a statistical procedure, which falls outside the scope of this paper.

2.2. Train compositions

The available rolling stock for a line usually consists of the same rolling stock type with different subtypes. A train can be composed of several train units of different subtypes, subsequently called a *composition*. Those subtypes differ in number of carriages and, as a result, in capacity, measured in numbers of first and second class passengers' seats. A train unit can be coupled to or uncoupled from a composition at the end of a trip. The position where units can be coupled to or uncoupled from the train is fixed for every station, i.e. at the rear end or at the front end of a train. In a terminal station, i.e. a station where trains arrive from and leave to the same direction (e.g. Vs) coupling and uncoupling usually occurs at the front end of an incoming train. In the station Rsd, where a train comes from the direction Vs and leaves into the direction Asd (or vice versa) a unit is coupled onto the front end of an incoming train and uncoupled from the rear end of an outgoing train. Therefore, not only the number of train units of each subtype, or the *combination*, is important, so is the order of the units in the composition, or the *permutation*. In addition, only one shunting operation for the same train can be performed at a station, i.e. coupling units onto or uncoupling units from the train but not

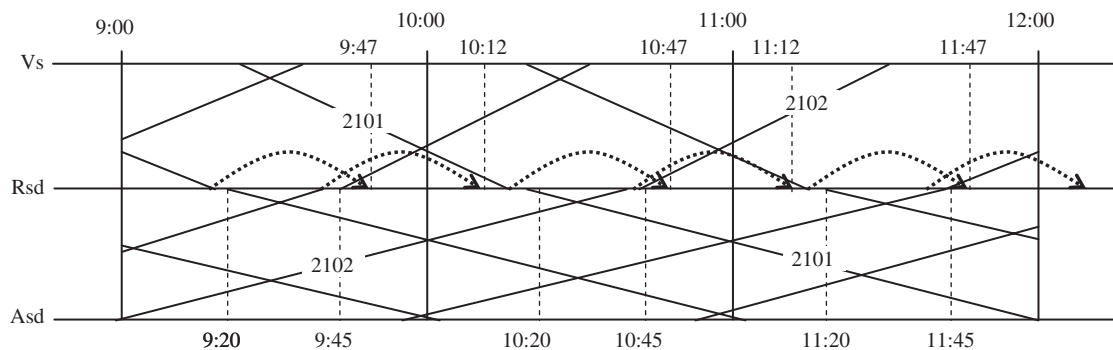


Fig. 3. Time-space diagram: Vs–Rsd–Asd.

both, because the time that a train stays at a station is too short to allow for several shunting operations. Therefore, it is not possible to remove a unit from the middle of the composition of a train. Finally, the model must also take into account that, if a train enters and leaves the station in the same direction, the composition of the train is turned around, i.e. the front unit becomes the rear one and vice versa. This occurs at the terminal stations Asd, Ddr, Gvc and Vs.

2.3. Inventory of train units at the stations

The limited availability of rolling stock causes interactions between the different trains. The initial inventory of train units must be divided over the stations. A train unit can only be coupled onto an incoming train if it is available in the right station and at the right moment in time. If a train unit is uncoupled, it is not immediately available to be coupled onto another train. A certain amount of “shunting time” must be taken into account. We further assume that a train unit is uncoupled at the moment that the train arrives in the station and a unit is coupled at the moment that the train departs from the station. Therefore, we determine the local shunting time at a station such that the arrival time increased with the shunting time of train A at the station is smaller than the departure time of train B at the station, if train B is the first train after the arrival of train A to which uncoupled train units from train A can be coupled. On the other hand, the arrival time of train A increased with the shunting time must be greater than the departure times of all trains leaving the station earlier than train B.

Fig. 3 presents the time-space diagram for the time window from 9:00 to 12:00, focusing on the trips arriving in and leaving from Rsd. The straight lines represent the trips, whereas the dotted arrows present the shunting time and indicate the moment in time that an uncoupled train unit becomes available to be coupled onto another train. The 2101 train for example leaves Vs at 9:26 and arrives in Rsd at 10:17. It proceeds its journey to Asd 3 minutes later at 10:20. Between 10:17 and 10:20, train units can be uncoupled from the train, but they are not immediately available to be coupled onto another train, due to the shunting time of 30 min in this example. This is reflected by the dotted arrow. Therefore, an uncoupled unit from the 2101 train cannot be coupled onto the 2102 train, coming from Asd and going to Vs at 10:45.

Finally, if the passenger demands and the timetables of consecutive days are very similar to each other, then it is desirable to operate the same rolling stock circulation over these consecutive days. A way of taking care of this is requiring that on such days the begin inventory at all stations for every rolling stock subtype must be equal to the end inventory. Within NS Reizigers, such a cyclicity constraint is usually imposed on the rolling stock circulations for Tuesdays, Wednesdays and Thursdays. However, such a cyclicity constraint may not be effective for days with a non-cyclic character, such as Mondays and Fridays. These days have a non-cyclic character since the passenger demand and the timetable during weekends are rather different from the passenger demand and the timetable during weekdays. For these non-cyclic days, a good circulation is such that at the end of the day the train units end in such stations that the trips of the next day can be served appropriately. However, in this paper, we assume that we have to take into account the cyclicity constraints per day.

2.4. Evaluation criteria

A circulation is evaluated based on three criteria, (i) the seat shortage kilometers, (ii) the number of shunting operations, and (iii) the carriage kilometers. Given the capacity of first and second class passengers of the subtypes of rolling stock and the composition of a train, the seat shortages on each subtrip can be computed. Those are multiplied by the length of the subtrip, and taking the sum over all subtrips results in the seat shortage kilometers for the entire trip. The second objective is minimizing the shunting operations, i.e. the number of times units must be (un)coupled to (from) a train. Those operations can cause delays, and since robustness is an important concern, their number should be controlled. The third objective is minimizing the total number of carriage kilometers, which are used as an approximation for the variable rolling stock cost. Carriage kilometers are positively correlated to power supply (electricity or diesel), but also to the maintenance of the rolling stock. After a certain number of kilometers, each train unit is directed to a maintenance station for a preventive check-up and possibly for a repair. Therefore, the maintenance cost is correlated with the number of carriage kilometers. Note that, as was mentioned earlier, maintenance planning and maintenance routing are *not* taken into account explicitly in the rolling stock circulation planning phase.

Since especially the first and second criterion cannot be expressed in an amount of money, we use weighting factors for the three criteria. By varying these weights, we obtain alternative solutions, which can be evaluated by the management of the railway operator or by the planners.

2.5. Literature review

In the literature, several related problems have been treated. The available literature witnesses the fact that rolling stock circulation problems exist in many variants, each with their own peculiarities. As far as we know, the only papers dealing with the circulation of *train units* are the following:

Schrijver [11] considers the problem of minimizing the number of train units of different subtypes for an hourly line of NS Reizigers, given that passengers' seat demand must be satisfied. The only restriction on the transition between two compositions on two consecutive trips is that the number of train units must be available. Other (un)coupling restrictions are ignored.

Ben-Khedher et al. [12] study the problem of allocating train units to the French High Speed Trains. Their rolling stock allocation system is based on a capacity adjustment model linked to the seat reservation system that seeks to maximize the expected profit.

Alfieri et al. [13] describe a model to determine the circulation of train units of *different* subtypes on a single line and day. This paper introduced the concept of a transition graph that is also used in the current paper. Their approach is based on an Integer Programming formulation that is solved with a commercial solver after a pre-processing step to reduce the sizes of the involved transition graphs. Their approach is tested on real-life instances of NS Reizigers.

Fioole et al. [14] describe a model for solving a rolling stock circulation problem that is very similar to the problem that we study in this paper. However, their model also allows for underway combining and splitting of trains, which happens regularly in the Netherlands. The model is solved by CPLEX. Since instances involving combining and splitting of trains may lead to long computation times, they also present several heuristic approaches based on fixing some of the binary variables.

Since many operators of passenger trains use locomotive hauled carriages, several papers study the circulation of locomotives and/or carriages. Brucker et al. [15] study the problem of finding a routing of carriages through a railway network, given a timetable. Since the compositions of the trains have been fixed a priori, they focus on the repositioning trips of the carriages from one station to another. Their solution approach is based on local search techniques like simulated annealing.

Cordeau et al. [16] present a Benders decomposition approach for the *locomotive and carriages* circulation problem. Their approach is based on the concept of *train consists*, i.e. a group of compatible units of equipment (locomotive(s), first and second class carriages) that travel along some part of the rail network. Computational experiments show that optimal solutions can be found in short computation times. In a subsequent paper, Cordeau et al. [17] extend their model with various real-life constraints, such as maintenance constraints.

Lingaya et al. [18] deal with the problem of assigning *carriages* to trains at VIA Rail in Canada. They present a model to adapt a master plan to additional information concerning the expected numbers of passengers. They allow for coupling and uncoupling of carriages at various locations in the network and explicitly take the order of the carriages

in the trains into account. Several other real-life constraints, such as maintenance requirements, are considered as well. Their solution approach is based on a Dantzig–Wolfe reformulation solved by column generation. Next, branch and bound is applied heuristically to obtain good integer solutions.

Abbink et al. [19] present a model to allocate different rolling stock types and subtypes of rolling stock to different lines. They present an Integer Programming model, that minimizes the seat shortages during the morning rush hours by allocating rolling stock types and subtypes with different capacities to all the trains running simultaneously at 8 o'clock, the busiest moment of the day. Their approach is applied to several instances of NS Reizigers that differ in the numbers of rolling stock types and subtypes that can be allocated to a line. For an excellent further survey on other optimization models for train routing and scheduling, we refer to Cordeau et al. (1998).

The main contribution of the current paper is the branch-and-price approach that we developed for solving the rolling stock circulation problem. As was indicated above, the problem studied in the current paper is very similar to the problem described by Fioole et al. [14]. However, as is shown by the computational results in Section 6, the branch-and-price approach described in the current paper is more powerful than the commercial solver CPLEX.

3. Model description

3.1. Transition graph

For each train, the possible changes in its composition at a station can be represented by a so-called *transition graph*, where the nodes represent the feasible train compositions on a trip and the arcs represent the feasible transitions between compositions at a station. Transition graphs were used also by Alfieri et al. [13]. Each train starts with an artificial trip from the depot at the departure station to the platform and ends with a trip to the depot at the arrival station of its last trip. The composition on these trips is the “0”-composition (the “empty” composition), i.e. all train units deployed on the first trip must be coupled to the train and removed from the inventory at the departure station, and all units deployed on the last trip must be added to the inventory at the final arrival station. For each trip, the expected numbers of first and second class passengers are given. Due to safety regulations, the expected seat shortage of passengers’ seats should not exceed a certain percentage of the available seats. In addition, there is a limit on the total length of a train due to the platform lengths of the stations where the train makes intermediate stops. Thus, many possible compositions can be excluded a priori.

Fig. 4 presents a transition graph of five trips for a train of the 2100–2400–2600 lines. The train compositions are presented in the format “ $U_r U_{r-1} \dots U_1$ ”, where U_r is the train unit at the rear end of the train, $U_{r-1} \dots U_2$ are the train units in the middle of the train, and U_1 is the train unit at the front end of the train. In this example, we assume that only train units of 4 and 6 carriages are available. The total length of a train cannot exceed 12 carriages and between Rsd and Vs the limit is just 9 carriages. The first real trip runs from Gvc to Asd and the possible compositions are given in Fig. 4. If composition “4” is used on this trip, then in Asd the composition can change to “44”, “46” or “444”. The compositions “4” and “6” are not allowed on the next trip between Asd and Rsd, because they do not have the required number of seats. Observe that the train composition is turned around in Asd such that composition “64” becomes “46”, etc. On the trip Rsd–Vs, only three compositions are possible, i.e. “4”, “6” and “44”, because the total length may not exceed 9 carriages. Suppose now that on this last trip composition “4” cannot be used, because the expected number of passengers is too high, then the composition “64” on the trip Asd–Rsd can be excluded as well, since it has no follower. The same holds then for composition “46” on Gvc–Asd. Next the train returns to Rsd and then to Asd, where composition “4” cannot be used because the expected number of passengers is too high.

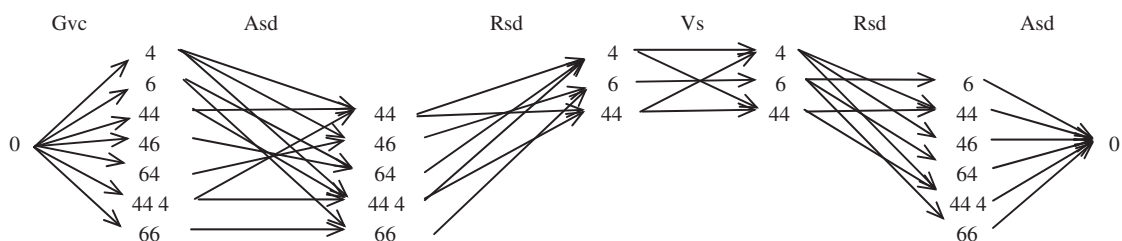


Fig. 4. Transition graph.

3.2. Inventory of train units

For each train, the model must select a composition for every trip. This can be considered as selecting a path in the transition graph from the “0”-node of the artificial first trip to the “0”-node of the artificial last trip. Since a train unit can only be coupled onto a train if it is available at the right time and station, the interaction between the different trains that run simultaneously is modelled by keeping track of the inventory position of each rolling stock subtype at all relevant events at the stations during the considered time period. The inventory position at an event e equals the initial allocation of rolling stock to the station augmented with the uncoupled train units and decreased with the coupled units before event e . The model must make sure that at any time, in all stations, the inventory positions of all rolling stock subtypes are non-negative. To this end, we determine the set of relevant events at the stations as follows. For every trip, we distinguish two events at the arrival station:

- (i) A Δ^- -event, corresponding to a potential decrease in the inventory position at the arrival station, and with the departure time of the following trip as realization time.
- (ii) A Δ^+ -event, corresponding to a potential increase in the inventory position at a station, and thus with the arrival time increased with the local shunting time as realization time.

The latter is required, because during the shunting operation the train unit cannot be coupled onto another train immediately, and should not immediately be added to the inventory at the station. After enumerating all events at a station, we sort them in increasing order of realization time. The *relevant* events are then the Δ^- -events that are immediately followed by a Δ^+ -event. Indeed, a Δ^+ -event as such is not relevant, since the previous inventory position must be non-negative and a Δ^+ -event can only increase the available rolling stock. As a result, it can never violate the condition that the inventory must be non-negative. If several Δ^- -events follow each other, then only the last Δ^- -event is relevant. Indeed, if after the last Δ^- -event the inventory position is still non-negative, then the latter holds for the Δ^- -events immediately preceding the last Δ^- -event as well. In the time–space diagram of Fig. 3, the realization times of the Δ^- -events are indicated below the diagram (9:20, 9:45, ..., 11:45) and the realization times of the Δ^+ -events are indicated above the diagram (9:47, 10:12, ...). The relevant events are then the Δ^- -events at 9:45, 10:45 and 11:45.

Finally, we note that train units, entering and leaving a station being part of the same train, are not added to nor removed from the inventory at the station at any time. We could add them upon arrival of the train and remove the same units when leaving again from the station. However, since these units do not really affect the inventory position, we ignore this temporary change in inventory.

3.3. Mathematical formulation

In order to formulate the rolling stock circulation problem, we use the following notation:

T	set of trips, index t ;
T_0	set of trips leaving from a depot, i.e. trips without a predecessor;
$p(t)$	trip preceding trip $t \in T \setminus T_0$;
M	set of different rolling stock subtypes, index m ;
N_m	the available number of train units of subtype $m \in M$;
K_t	set of possible compositions for trip $t \in T$, indices j or k ;
$P_{t,k}$	set of possible preceding compositions if composition $k \in K_t$ is used on trip $t \in T$;
$F_{t,k}$	set of possible following compositions if composition $k \in K_t$ is used on trip $t \in T$;
S	set of stations, index s ;
E_s	set of relevant events at station s , i.e. the initial state “0”, a Δ^- -event if it is immediately followed by a Δ^+ -event, and the final state “ $\mathcal{F}$ ”, index e ;
$q(e)$	relevant event at station s immediately preceding event $e \in E_s$.

In our model, we use two types of decision variables. First, we associate a decision variable with every arc in the transition graphs of the trains, indicating whether or not the arc is used. More specifically, for every trip t , composition $k \in K_t$ and allowed following composition $j \in F_{t,k}$, a variable $x_{t,k,j}$ is defined, which equals 1 if composition k

is used on trip t and composition j is used on the next trip, and 0 otherwise. With every variable, a cost parameter $c_{t,k,j}$ is associated, which equals the weighted sum of the seat shortage kilometer cost, the carriage kilometer cost if composition k is used on trip t , and the shunting cost if composition k differs from composition j on the next trip. In order to establish the link between the arc variables and the inventory positions at the stations, we associate with every variable two parameters, indicating the number of train units of a certain subtype that is (un)coupled:

- $\alpha_{t,k,j}^m$ the number of uncoupled units of subtype m at the end of trip t , given that composition k changes to composition j ;
 $\beta_{t,k,j}^m$ the number of coupled units of subtype m at the end of trip t , given that composition k changes to composition j .

For the trips $t \in T_0$, the number of coupled train units $\beta_{t,k,j}^m$ of subtype m equals all the units of the composition j , because the composition k on the artificial trip is the “0”-composition (see Fig. 4). For final trips t to the depot, the number of uncoupled units $\alpha_{t,k,j}^m$ of subtype m equals all the units of composition k , since in this case composition j is again the “0”-composition.

A second type of decision variables consists of inventory variables at the stations. With every relevant event $e \in E_s$ at station s and for every subtype $m \in M$, we associate a decision variable $I_{e,s}^m$, indicating the number of units of subtype m at station s immediately after event e . In order to model the changes in the inventories of the subtypes at a station s between two relevant events, $q(e)$ and $e \in E_s$, we must determine the set of trips corresponding to a Δ^- -event, causing a decrease in the inventory, and the set of trips corresponding to a Δ^+ -event, causing an increase in the inventory. Therefore, let r_e be the realization time of relevant event e , then we can define both sets as follows:

- $A_{e,s}$ the set of trips, arriving in station s , for which the arrival time increased with the shunting time at station s belongs to the interval $(r_{q(e)}, r_e]$ with $e \in E_s$;
 $D_{e,s}$ the set of trips, arriving in station s , for which the departure time of the following trip at station s belongs to the interval $(r_{q(e)}, r_e]$ with $e \in E_s$.

Observe that the set $D_{e,s}$ does not contain trips *departing* from station s , but trips *arriving* in station s . In order to determine whether or not a particular trip belongs to the set, however, we use the departure time of the next trip, since this is the realization time of a Δ^- -event.

In the example of Fig. 3, we recall that the relevant events at the station Rsd occur at 9:45, 10:45 and 11:45, and we consider the time interval (10:45, 11:45]. Then the set $D_{e,s}$ consists of two trips: (1) Vs–Rsd arriving in Rsd at 11:17 and (2) Asd–Rsd, arriving in Rsd at 11:42. The set $A_{e,s}$ consists of two trips as well: (1) Vs–Rsd arriving in Rsd at 10:17 and (2) Asd–Rsd, arriving in Rsd at 10:42. Now, we can formulate the problem as follows:

(i) Minimize the cost:

$$\min \sum_{t \in T} \sum_{k \in K_t} \sum_{j \in F_{t,k}} c_{t,k,j} x_{t,k,j} \quad (1)$$

subject to

(ii) Flow conservation constraints:

$$\sum_{k \in K_t} \sum_{j \in F_{t,k}} x_{t,k,j} = 1 \quad \forall t \in T_0, \quad (2)$$

$$\sum_{j \in P_{t,k}} x_{p(t),j,k} = \sum_{j \in F_{t,k}} x_{t,k,j} \quad \forall t \in T \setminus T_0 \quad \forall k \in K_t. \quad (3)$$

(iii) Initial inventory:

$$\sum_{s \in S} I_{0,s}^m = N_m \quad \forall m \in M. \quad (4)$$

(iv) Inventory constraints:

$$I_{q(e),s}^m + \sum_{t \in A_{e,s}} \sum_{k \in K_t} \sum_{j \in F_{t,k}} \alpha_{t,k,j}^m x_{t,k,j} - \sum_{t \in D_{e,s}} \sum_{k \in K_t} \sum_{j \in F_{t,k}} \beta_{t,k,j}^m x_{t,k,j} - I_{e,s}^m = 0$$

$$\forall s \in S \quad \forall m \in M \quad \forall e \in E_s \setminus \{0\}. \quad (5)$$

(v) Cyclicity constraints:

$$I_{0,s}^m = I_{f,s}^m \quad \forall s \in S \quad \forall m \in M. \quad (6)$$

(vi) Non-negativity and integrality;

$$x_{t,k,j} \in \{0, 1\} \quad \forall t \in T \quad \forall k \in K_t \quad \forall j \in F_{t,k}, \quad (7)$$

$$I_{e,s}^m \geq 0 \quad \forall s \in S \quad \forall m \in M \quad \forall e \in E_s. \quad (8)$$

Constraints (2) impose that for every trip without a predecessor exactly one arc must be chosen, resulting in a composition for the first trip and the transition in composition at the arrival station of the first trip. Constraints (3) are flow conservation constraints, i.e. for a node in the transition graph characterized by a trip t and a composition $k \in K_t$, the incoming flow, which equals the sum of the flows of the arcs arriving in the node, must be equal to the outgoing flow. Constraints (4) impose that the sum of the initial inventories at the stations must be equal to the total availability N_m for each subtype $m \in M$. Constraints (5) are the balancing constraints, stating that the inventory of subtype $m \in M$ at station $s \in S$ of the previous event increased with the number of uncoupled train units in the time period between event $q(e)$ and event e and decreased with the number of coupled train units of subtype m in the time period between event $q(e)$ and event e must be equal to the inventory at event e . The cyclicity constraints (6) impose that the initial inventory at a station must be equal to the end inventory for every subtype. Finally, constraints (7) specify the integrality condition on the arc variables, whereas constraints (8) reflect the non-negativity condition for the inventory positions.

4. Dantzig–Wolfe decomposition

4.1. Formulation

The rolling stock circulation problem (1)–(8) is suitable to apply Dantzig–Wolfe decomposition [20]. In contrast to multicommodity flow decomposition approaches in the literature (e.g. [9,10]), where paths through the network are generated for each commodity and the capacity constraints are considered in the master problem, we decompose the problem based on the trains. Each train can be composed of several commodities, i.e. the rolling stock subtypes. In the Dantzig–Wolfe reformulation, we associate a variable with each path through the transition graph of a train. The number of trains equals the number of trips without a predecessor. That is, the number of trips in the set T_0 . To state the Dantzig–Wolfe reformulation, we need the following additional notation:

- R set of trains, index i ;
- Ω^i set of paths through the transition graph of train $i \in R$, index ω ;
- T^i set of trips corresponding to train i .

Again, we introduce two types of decision variables, i.e. (i) the inventory variable $I_{e,s}^m$ for every relevant event $e \in E_s$ at station s and for every subtype $m \in M$ as defined before, and (ii) the path variable λ_{ω}^i , which equals 1 if path $\omega \in \Omega^i$ is selected in the solution, and 0 otherwise. The cost associated with path $\omega \in \Omega^i$, denoted as χ_{ω}^i , equals the cost of the arcs used by the path. Finally, with every path $w \in \Omega^i$, we associate a set of parameters $q_{t,k,j}^w$ for every trip $t \in T^i$ and for every composition $k \in K_t$ and following composition $j \in F_{t,k}$. These parameters are associated with the arcs in the transition graph for the train $i(t)$ covering trip t , and describe the path $\omega \in \Omega^i$. Therefore, the parameter $q_{t,k,j}^w$

equals 1 if path ω uses the corresponding arc in the transition graph, and 0 otherwise. Now the Dantzig–Wolfe reformulation can be stated as follows:

(i) Minimize the cost:

$$\min \sum_{i \in R} \sum_{\omega \in \Omega^i} \lambda_{\omega}^i \lambda_{\omega}^i \quad (9)$$

subject to

(ii) Select a path for every train:

$$\sum_{\omega \in \Omega^i} \lambda_{\omega}^i = 1 \quad \forall i \in R. \quad (10)$$

(iii) Initial inventory constraints (4):

(iv) Inventory constraints:

$$I_{q(e),s}^m + \sum_{t \in A_{e,s}} \sum_{\omega \in \Omega^{i(t)}} \sum_{k \in K_t} \sum_{j \in F_{t,k}} \alpha_{t,k,j}^m q_{t,k,j}^{\omega} \lambda_{\omega}^{i(t)} - \sum_{t \in D_{e,s}} \sum_{\omega \in \Omega^{i(t)}} \sum_{k \in K_t} \sum_{j \in F_{t,k}} \beta_{t,k,j}^m q_{t,k,j}^{\omega} \lambda_{\omega}^{i(t)} - I_{e,s}^m = 0$$

$$\forall s \in S \quad \forall m \in M \quad \forall e \in E_s \setminus \{0\}. \quad (11)$$

(v) Cyclicity constraints (6):

(vi) Non-negativity and integrality (8):

$$\lambda_{\omega}^i \in \{0, 1\} \quad \forall i \in R \quad \forall \omega \in \Omega^i. \quad (12)$$

Constraints (10) impose that for every train a path must be selected and are known as the *convexity* constraints. Constraints (11) replace the original inventory constraints (5). In general, this formulation has a huge number of path variables, and it is impossible to enumerate them all. Therefore, we solve the LP relaxation of problem (4), (6), (9)–(12) using column generation.

4.2. Solving the LP relaxation

The problem (4), (6), (9)–(11) is initialized with a small subset of the columns from the sets Ω^i , $i \in R$ and is called the *restricted master problem*. This problem is then solved using the simplex algorithm. Based on the dual prices corresponding to constraints (11), denoted as $\pi_{e,s}^m$, new columns are generated from the sets of paths Ω^i , $i \in R$. Our goal is to find a path with minimum reduced cost for each train $i \in R$. The reduced cost consists of four components, namely (i) the seat shortage cost, (ii) the shunting cost and (iii) the carriage kilometer cost, as explained before, and (iv) the rolling stock cost or revenue. Indeed, the dual prices $\pi_{e,s}^m$ can be interpreted as the value of one train unit of subtype m in station s during the time interval $(r_{q(e)}, r_e]$, i.e. if for a train that departs from station s during the time interval $(r_{q(e)}, r_e]$, b units of subtype m are coupled onto the train, then an additional cost of $b \times \pi_{e,s}^m$ is incurred. Similarly, if a units of subtype m are uncoupled from the train and added to the inventory during the period $(r_{q(e)}, r_e]$, then a revenue is received of $a \times \pi_{e,s}^m$.

More formally, the pricing problem P for train i consists of finding the shortest path through the transition graph of train i . The costs of the arcs are modified based on dual price information of the last solved restricted master problem. Let $a(t)$ be the arrival station of trip t , and let $e^+(t)$ be the first relevant event after the Δ^+ -event of trip t in the station, i.e. $t \in A_{e^+(t),a(t)}$. Similarly let $e^-(t)$ be the relevant event at the arrival station of trip t after or corresponding to the Δ^- -event of trip t , i.e. $t \in D_{e^-(t),a(t)}$. Then the pricing problem for train i can be stated as follows:

Minimize the reduced cost:

$$z_P^i = \min \sum_{t \in T^i} \sum_{k \in K_t} \sum_{j \in F_{t,k}} \left(c_{t,k,j} - \sum_{m \in M} (\pi_{e^+(t),a(t)}^m \alpha_{t,k,j}^m - \pi_{e^-(t),a(t)}^m \beta_{t,k,j}^m) \right) x_{t,k,j} \quad (13)$$

subject to

$$\sum_{k \in K_t} \sum_{j \in F_{t,k}} x_{t,k,j} = 1 \quad \forall t \in T_0 \cap T^i, \quad (14)$$

$$\sum_{j \in P_{t,k}} x_{p(t),j,k} = \sum_{j \in F_{t,k}} x_{t,k,j} \quad \forall t \in T^i \setminus T_0 \quad \forall k \in K_t. \quad (15)$$

Observe that problem (13)–(15) is a shortest path problem. As a result, it can be solved in polynomial time. Columns are added to the master problem as long as columns with negative reduced cost exist, i.e. $z_p^i - \rho_i < 0$, where $\rho_i, i \in R$ denotes the dual price of the convexity constraint (10). However, due to numerical instability, it may occur that columns price out, i.e. have a negative reduced cost, although they are already included in the restricted master problem and, as a result, convergence is not assured. To cope with this problem, we use a somewhat different stopping criterion. At each column generation iteration, we solve the pricing problem (13)–(15) for every train. Let $\bar{Z}_{LP}$ be the optimal LP objective function value of the restricted master problem, and let Z_{LP} be the optimal LP value of model (4), (6), (9)–(11), then the following inequality holds (see e.g. [21]),

$$\bar{Z}_{LP} \geq Z_{LP} \geq \bar{Z}_{LP} + \sum_{i \in R} (z_p^i - \rho_i). \quad (16)$$

As a result, at each iteration, we have a lower bound and an upper bound on the optimal LP objective function value Z_{LP} . We stop generating columns if the relative difference between the lower and upper bound is less than ε (where $\varepsilon = 5 \times 10^{-5}$ in our numerical experiments), or

$$\sum_{i \in R} (z_p^i - \rho_i) / \left(\bar{Z}_{LP} + \sum_{i \in R} (z_p^i - \rho_i) \right) \leq \varepsilon. \quad (17)$$

As long as condition (17) is not met, the pricing problem is solved for every train and we look for the generated columns with the most negative reduced cost over all trains, denoted as c^* . Next, we add all the generated columns for which $z_p^i - \rho_i < \delta \times c^*$ to the restricted master (where $\delta = 0.01$ in our numerical experiments). We do this in order to avoid the possibility that columns with a slightly negative reduced cost are added to the restricted master problem if stopping criterion (17) is not met yet, because for other trains new columns with a relatively high negative reduced cost still price out. As a result, the procedure is not sensitive to the choice of the parameter δ .

It is well-known that columns generated in earlier stages of the column generation procedure are often of poor quality. This is because the initial dual prices are far from the optimal ones. Therefore, a “good” initialization of the restricted master problem is useful. However, we propose a two phase procedure to cope with this issue. In a first phase, we restrict the solution space of the pricing problems using problem-specific knowledge. For example, in our problem, it seems unlikely that, in an optimal solution, train units are coupled to (uncoupled from) a train if on the next trip the expected number of passengers is lower (higher). Therefore, we initially forbid those transitions in the transition graphs of the trains. Next, we apply the column generation procedure until condition (17) is met. In a second phase, we allow all feasible transitions and apply the column generation procedure again. This two phase procedure considerably speeds up the convergence of the column generation algorithm. In our computational experiments, we observe a decrease in computation time for the LP relaxation of a factor 5–7 on average for larger instances compared to a single phase procedure.

Finally, since the column generation subproblem can be solved as a shortest path problem, it has the so-called *integrality property* [22]. As a consequence, the column generation approach leads to the same lower bound as the LP relaxation of problem (1)–(8).

5. Branch and price

After solving the LP relaxation, branching is needed to obtain a proven optimal integer solution. Since for each $i \in R$ only a subset of the paths from Ω^i is inserted in the master problem for solving the LP relaxation, and not necessarily all paths required for the optimal integer solution, the column generation procedure must be applied at every node of the branch-and-price tree with an appropriately modified pricing problem. These modifications to the pricing problem

reflect the branching decisions taken at the current node, and the nodes higher in the tree on the path from the root node to the current node.

Our branching scheme consists of imposing and forbidding the deployment of a set of compositions on certain trips. More specifically, we build a binary search tree, where at the first node we impose that for a trip t a composition must be chosen from a certain subset C_t of the compositions K_t , and at the twin node a composition from the set $K_t \setminus C_t$ must be selected. To that end, let $q_{t,k}^\omega = \sum_{j \in F_{t,k}} q_{t,k,j}^\omega$. As a result, $q_{t,k}^\omega$ equals 1 if trip t uses composition k in path ω , and 0 otherwise. Then the branching constraints are given by the following expression:

$$\sum_{\omega \in Q^{i(t)}} \sum_{k \in C_t} q_{t,k}^\omega \lambda_\omega^{i(t)} = 1 \quad \text{and} \quad \sum_{\omega \in Q^{i(t)}} \sum_{k \in C_t} q_{t,k}^\omega \lambda_\omega^{i(t)} = 0. \quad (18)$$

The branching constraints (18) need not be added explicitly to the master program. Indeed, they can be implemented by deleting the appropriate columns from the master program (4), (6), (9)–(11) and by avoiding that these columns price out again by deleting the nodes corresponding to forbidden compositions from the transition graphs. Furthermore, we use a *best first* search strategy. That is, we always branch from the node with the lowest lower bound. Ties are broken randomly.

We now discuss the sufficient conditions to have an integer solution, and next how we select the branching constraints. First, we argue that it is sufficient to impose that a *combination*, i.e. the number of train units of each subtype, neglecting the order, is selected for each trip, and not necessarily a *permutation* in order to obtain an integer solution. Suppose for example that, in the transition graph of Fig. 4, we have imposed that on the trip between Asd and Rsd, the combination consisting of one unit of subtype “4” and one unit of subtype “6” must be used, corresponding to the compositions or permutations “46” and “64”. Then, on the next trip between Rsd and Vs only two combinations and permutations can be used, i.e. the composition that follows “64” namely “4” and the follower of “46”, namely “6”. As a result, imposing a branching constraint for this following trip, e.g. use combination “4”, has the consequence that for the previous trip permutation “64” must be used. On the other hand, at the twin node (i.e. forbid combination “4”), combination “6” must be used on the Rsd–Vs trip and “46” between Asd and Rsd. In general, by branching on combinations, all fractional permutations can be eliminated as well, and the path through the transition graph is uniquely determined.

However, there is one exception. It can happen that there exist distinct paths through the transition graph of a train, although the combination is fixed for every trip. This is the case if a train consists of a core component of at least two units of a different subtype that is deployed on all trips of the train during the day. Suppose, for instance, that a train consists of at least one unit of subtype “4” and one unit of subtype “6”. During the day, other units can be coupled to and later uncoupled from this core component, to which two permutations correspond, i.e. “46” and “64”. As a result, whatever train units are coupled or uncoupled, two paths through the transition graph do exist. In this case we can arbitrarily select one of those paths and consider it as integer, since both are feasible and the relation with the other trains is identical for both paths, because the same units are (un)coupled after every trip. This insight allows to branch by imposing that a combination must be used for each trip rather than a permutation, which results in a smaller search tree and avoids symmetry.

The selection of the trip for which we impose a branching constraint is crucial in order to obtain an integer solution quickly. Based on the current master solution, we determine the fraction that a particular combination is used for each trip. From the set of trips for which there exists a fractional combination, we select the trip with the highest number of passengers and the combination with the largest fraction. The set of compositions C_t of branching constraint (18) are then all permutations that correspond to the selected combination.

Finally, throughout the branch-and-price tree, we compute at every column generation iteration the lower bound on the LP value of Eq. (16). If this is greater than the best IP solution found, then we can stop generating columns and prune the current node in our branch-and-price tree.

6. Numerical experiments

In order to test the described procedure, we carried out several computational experiments. The experiments are performed on an IBM NetVista 6343-25G Pentium 4 1.6 GHz PC, using the Windows 98 operating system. Our algorithm is coded in C++ and linked with the Extended LINDO/PC optimization library release 6.1 for solving the LP subproblems.

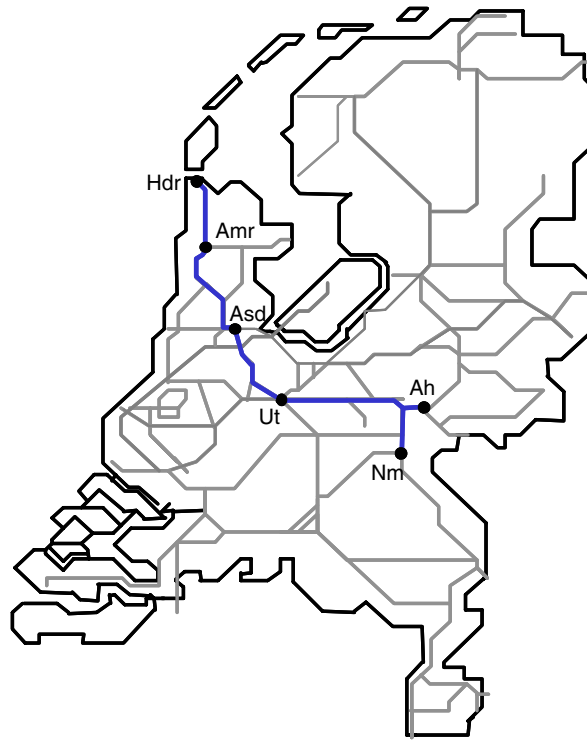


Fig. 5. The 3000 line.

Table 3
Data sets

Line	Trips	Trains	Stations
2100–2400–2600	182	15	5
3000	115	12	2

6.1. Description of the data sets

Our procedure is tested on two cases, i.e. the 2100–2400–2600 lines presented in Table 1 and depicted in Fig. 2, and the 3000 line, that runs between Nijmegen (Nm) and Den Helder (Hdr) (see Fig. 5).

A 3000 train goes from Hdr to Alkmaar (Amr), where train units can be coupled or uncoupled. Then the train continues to Amsterdam (Asd), Utrecht (Ut), Arnhem (Ah) and Nijmegen. In Arnhem the train leaves the station in the same direction as it arrives. As a result, the composition is turned around such that the first train unit becomes the last and vice versa. This can be taken into account in the transition graphs, as explained earlier. In the terminal station Nm, train units can be coupled or uncoupled. As a result, there are two stations where shunting operations can occur, i.e. Nm and Amr. Table 3 gives the number of trains, the number of trips that must be performed during one day, and the number of stations where (un)coupling can occur during the day. Further details of the 3000 line can be found in Alfieri et al. [13].

In all cases, we consider the rolling stock circulation problem for a single generic weekday. We assume that the cyclicity constraints (6) on the inventories have to be respected. There exist three subtypes of train units, i.e. double deck units of three, four or six carriages. These are called DD3, DD4 and DD6 in the following. However, we also did experiments with DD4 and DD6 units only. The maximum number of carriages on each trip is 12, except for the trip Rsd–Vs where it is 10, and between Vs–Rsd, Amr–Hdr, and Hdr–Amr, where it is 9. For both series, we solve several instances by varying the availabilities of the subtypes of the train units and by varying the weights of the objective function.

Table 4
Computational results

$w_s = 0$	2000		3000	
	2 (62)	3 (82)	2 (45)	3 (58)
T_{IP}	5.86	91.27	0.70	10.04
T_{LP}	1.32	14.21	0.35	1.13
Nds	11.10	9.88	10.89	17.52
K_{LP}	304.61	574.34	153.67	263.48
K_{IP}	401.32	860.74	184.11	443.69
Nc	145.47	155.49	124.24	127.69
Gap (%)	0.92	0.68	0.34	0.86
CKM	127,171	133,393	113,671	118,617
S_1	14.61	0.00	104.49	24.21
S_2	49,095	23,980	142,962	143,120
SHT	78.68	77.12	70.71	68.10

Table 5
Computational results

$w_s = 5$	2000		3000	
	2 (62)	3 (82)	2 (45)	3 (58)
T_{IP}	4.49	87.69	0.60	9.84
T_{LP}	1.19	13.02	0.36	1.17
Nds	9.81	12.93	5.07	17.69
K_{LP}	305.11	566.82	166.82	279.88
K_{IP}	398.06	870.70	197.44	462.24
Nc	145.47	155.49	124.24	127.69
Gap (%)	0.90	0.67	0.34	0.86
CKM	133,231	138,878	117,074	120,520
S_1	11.81	0.00	104.49	24.21
S_2	49,101	23,980	142,962	143,120
SHT	53.39	55.39	53.18	56.69

6.2. Computational results

Tables 4 and 5 present the computational results. In Table 4, the weight given to a shunting operation, denoted by w_s , equals 0, whereas in Table 5 this weight equals 5. The weight given to a seat shortage kilometer equals 1 for a seat shortage kilometer in the second class (w_2), and 2 for a seat shortage kilometer in the first class (w_1). The weight of a carriage kilometer (w_{car}) equals 0.01.

The column “2000” (“3000”) gives the results for the lines 2100–2400–2600 (3000). The second row gives the number of subtypes, i.e. 2 or 3, and the number in brackets gives the number of instances solved for this case, e.g. for the 3000 line and three subtypes, we solve 58 instances, by varying the availabilities of the different subtypes. The remaining rows have the following meaning:

T_{IP}	average CPU time to find the optimal integer solution (in seconds);
T_{LP}	average CPU time to find the LP relaxation at the root node (in seconds);
Nds	average number of nodes in the branch-and-price tree;
K_{LP}	average number of generated columns at the root node;
K_{IP}	average total number of generated columns;
Nc	average number of carriages;
Gap	average relative gap between the LP bound at the root node and the optimal solution (that is, the difference in IP solution and LP relaxation at the root node divided by the latter);
CKM	average number of carriage kilometers;

- S_1 average number of first class seat shortage kilometers;
- S_2 average number of second class seat shortage kilometers;
- SHT average number of shunting operations (i.e. coupling and uncoupling).

Based on Tables 4 and 5 we conclude that our procedure can solve all instances to optimality in rather short computation times. If the number of subtypes equals 2, then our algorithm requires on average only 6 s to obtain proven optimal solutions for the 2100–2400–2600 lines and less than a second for the 3000 line. If the number of subtypes equals 3, then the required CPU time increases to about 90 s for the 2100–2400–2600 lines and about 10 s for the 3000 line.

As the pricing problems are relatively small shortest path problems, the computation time is almost entirely spent on solving the master LPs. The low average gap indicates that the LP lower bound is close to the optimal integer solution and, therefore, only few nodes must be investigated to obtain a proven optimal solution. We also did experiments with different weights assigned to the cost components. Since the computation time turned out to be insensitive to these changes, we opt for not reporting the corresponding results.

6.3. Comparison with CPLEX and with practice

In this section, we compare the computation times of our branch-and-price procedure to those that are obtained when formulation (1)–(8) is solved by commercial software, in particular the modelling language OPL Studio 3.6 using the solver CPLEX 8.0. We compare our branch-and-price procedure for the same 58 instances of the 3000 line with shunting weight equal to 0 and with three different subtypes. We make sure that a similar branching strategy is used for formulation (1)–(8) as for the branch-and-price algorithm. To do so, we first enforce integrality on the combinations of train units deployed on a trip. Second branching priority is given to trips with most passengers.

To implement the branching on combinations, auxiliary binary variables, one for each combination that can be deployed on a trip, are introduced in the formulation. The relation between the permutations and the combinations is enforced by equality constraints. To be more precise, let the decision variable $w_{t,c}$ be 1 if combination c is deployed on trip $t \in T$, and 0 otherwise. Let Π_c be the set of permutations corresponding to combination c . Then, for every trip and possible combination, the following constraints are added to the problem: $w_{t,c} = \sum_{k \in \Pi_c} \sum_{j \in F_{t,k}} x_{t,k,j}$. These constraints state that, if a combination on a trip is selected in the solution, then a corresponding permutation must be chosen as well. It is then sufficient to declare the combination variables $w_{t,c}$ as integer and the permutation variables as fractional. As a result, the branching strategy for both models is comparable, and, hence, the numbers of investigated nodes are on average almost equal.

The average CPU-time, however, is about 10 s for the branch-and-price algorithm (see Table 4) versus 227 s for CPLEX. This difference is mainly caused by the fact that we use a dedicated shortest path algorithm for quickly solving the pricing problems in the transition graphs. As was mentioned earlier, the computation time is almost entirely spent on the master LPs. Moreover, CPLEX must solve relatively large LP models at every node of the branch-and-bound tree, whereas the branch-and-price algorithm has to solve a large number of relatively small models. The remaining part of the time difference can be explained by the overhead involved in OPL Studio.

For the 2100–2400–2600 lines, we compared the circulation currently (2004) used in practice (employing 1 DD3, 16 DD4's and 12 DD6's), with the circulations obtained by our model. The results of four scenarios are presented in Table 6. In the first two scenarios, we emphasize the minimization of the seat shortage kilometers and use the same weights as for the experiments of Tables 4 and 5. In the next two scenarios, we give a higher weight to the carriage kilometers. We observe that the second class seat shortage kilometers in practice are much higher than in the circulations proposed by our model. If we emphasize the carriage kilometers, we are able to find circulations with less carriage kilometers and with about 65% less second class seat shortage kilometers. We can conclude that our procedure is reasonably fast to perform several kinds of what-if studies.

6.4. Further what-if studies

In this section, we present two further what-if studies. For the 2100–2400–2600 lines, we first investigate the effect of using only DD4 and DD6 units on these lines. We solve the circulation problem for every combination of numbers of

Table 6
Comparison between circulation in practice and solution of the model

		Seat shortages ^a		Carriage km ^b	
Emphasis		$w_s = 0$	$w_s = 5$	$w_s = 0$	$w_s = 5$
Practice					
CKM	127,521	130,906	135,084	125,321	127,860
S_1	0	0	0	0	0
S_2	90,572	23,583	23,583	29,981	27,737
SHT	60	77	58	76	62

^a $w_1 = 2, w_2 = w_1/2, w_{car} = 0.01$.
^b $w_1 = 1/30, w_2 = w_1/2, w_{car} = w_1$.

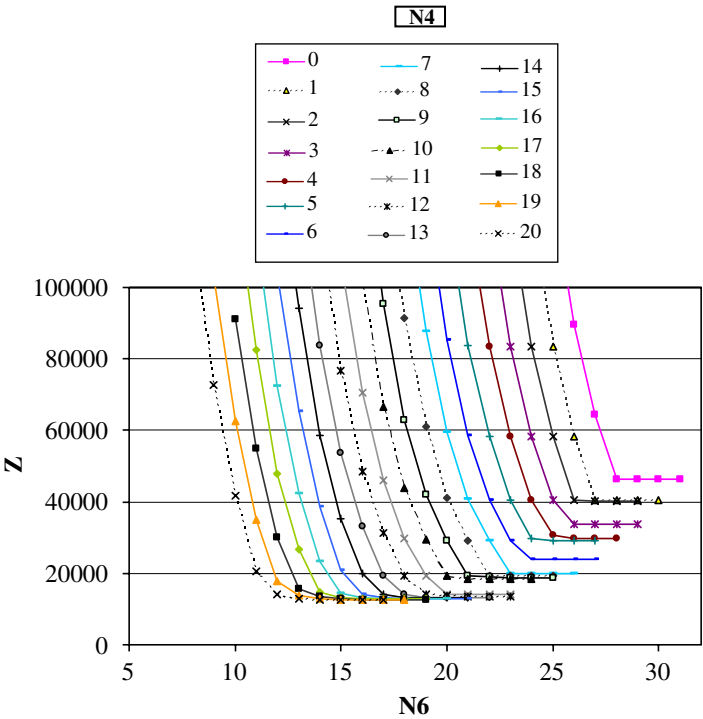


Fig. 6. Circulation cost for the 2100–2400–2600 line (only units of 4 and 6 carriages).

DD4 and DD6 units for which the total number of carriages is between 120 and 180. A lower availability of carriages results in an infeasible problem and a higher availability in excess capacity.

Fig. 6 presents the results of this analysis. On the Y-axis the total cost is given and on the X-axis the number of available DD6's. Every line on the chart presents the changes in the cost for a fixed number of DD4's. One clearly observes that the marginal profit of one additional DD6 sharply decreases and quickly becomes 0. For example, for 3 DD4's, more than 26 DD6's cannot be deployed efficiently. We also observe that for the same number of carriages, the solution becomes more efficient if the share of DD4's increases. For example for 160 carriages, the total cost for 4 DD4's and 24 DD6's is around 40,000, whereas the cost for 7 DD4's and 22 DD6's is about 30,000, and even less than 20,000 for 10 DD4's and 20 DD6's. Obviously, such results could have been guessed a priori, since units of DD4 are more flexible than units of DD6, because of their shorter length.

In a further what-if study, we compare in Table 7 the results for the standard situation for the 3000 line, called (i) baseline, with the results obtained by two relaxations: (ii) allowing coupling and uncoupling (shunting) in Hdr, and (iii) increasing the maximum train length from 9 to 12 carriages between Amr and Hdr and vice versa. In case (ii), the number of trips increases then from 115 to 148.

Table 7
Cases for the 3000 line

	M	Case	CKM	S_1	S_2	SHT	T_{IP}
$w_s = 0$	2	(i) Baseline	113,671	104.5	142,962	70.7	0.70
		(ii) Shunting in Hdr	112,526	104.5	142,933	79.0	2.29
		(iii) Max 12	114,084	97.5	125,505	65.6	0.73
	3	(i) Baseline	118,617	24.2	143,120	68.1	10.04
		(ii) Shunting in Hdr	117,934	24.2	143,044	71.5	62.46
		(iii) Max 12	120,123	28.6	125,206	61.7	12.27
$w_s = 5$	2	(i) Baseline	117,074	104.5	142,962	53.2	0.60
		(ii) Shunting in Hdr	116,944	104.5	142,933	53.3	1.64
		(iii) Max 12	119,048	97.5	125,505	42.4	0.67
	3	(i) Baseline	120,521	24.2	143,120	56.7	9.84
		(ii) Shunting in Hdr	120,462	24.2	143,044	56.8	61.13
		(iii) max 12	123,722	28.6	125,206	43.7	16.95

The column labelled M gives the number of subtypes. The entries in the table are the averages over the same instances as in Tables 4 and 5. The average computation time for case (ii) increases to about 1 min for the instances with 3 subtypes, due to the higher number of trips. We observe that there is no effect on the average first class seat shortage kilometers and that the effect on the second class seat shortage kilometers is insignificant, i.e. about 0.02% for $M = 2$ and 0.05% for $M = 3$. The reduction in carriage kilometers is very small if a positive weight is given to the shunting operations. If not, a reduction of 1% can be achieved if two subtypes are used and 0.5% if three subtypes are used. As a result, an additional shunting possibility in Hdr does not have a big impact on the solution quality. Increasing the maximum train length to 12 carriages, however, results in a decrease of the seat shortage kilometers with about 10% and an increase in carriage kilometers with about 2%. It can be concluded that the limited maximum train length of 9 carriages between Amr and Hdr is indeed a bottleneck, that could be solved by increasing the lengths of the platforms there.

7. Conclusions and future research

In this paper, we presented a model to determine an efficient rolling stock circulation for a set of interacting railway lines. We developed a branch-and-price algorithm to solve the problem to proven optimality. The algorithm was tested on real-life instances of NS Reizigers, the main Dutch operator of passenger trains. The required CPU time is short, as follows from a comparison with CPLEX 8.0. Therefore, what-if analyses can be performed easily. The model can be a useful tool for planners to determine the number of subtypes allocated to a line. In addition, the circulation proposed by the model can be a good starting point to make the real circulation plan. However, this will require the availability of a user-friendly graphical user-interface in addition to our algorithmic approach.

In our future research, we will focus on the application of the proposed procedure to the case of underway splitting and combining of trains, as described by Fioole et al. [14]. We will aim at expanding the branch-and-price approach to cover this issue as well. A first investigation has shown that the branch-and-price algorithm can handle this issue, as long as each pair of trains that originated from splitting a single train is later combined with each other to form a single train again. In that case the concept of the transition graphs can still be applied. However, the application of the branch-and-price algorithm is more complicated if there are split trains that are combined with other trains than the trains that they were split from earlier. In such cases, which occur regularly in the Dutch practice, there are additional interactions between the corresponding transition graphs. This is a subject for further research. Also other complicating issues, such as a larger number of lines per instance and a larger number of types and subtypes per line is a subject for further research.

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